Math 196 Lecture Notes: Week 8

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0.1 The Determinant in Dimension Two

For a 2×2 matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

we have seen that A is invertible if and only if $\det(A) = ad - bc \neq 0$. We also saw how to interpret this quantity as the area of the parallelogram with sides given by the columns of A, the image of the unit square under A.

0.2 Desiderata for the Determinant in General

For a general $n \times n$ matrix A, we will define $\det(A)$ in a way that generalizes this. In particular:

1. $\det(A)$ is a polynomial in the entries of A with coefficients ± 1.
2. A is invertible if and only if $\det(A) = 0$.
3. $|\det(A)|$ represents the volume of the image of the n-dimensional unit cube under A, and the sign of $\det(A)$ tells us whether A preserves or reverses orientation (whatever this means).

I will give you a definition of $\det(A)$ of the form specified in (1), and we will see why (2) is true later when we connect this definition to row operations. I will not say much more about (3) in higher dimensions.

1 Determinants in General

1.1 The Determinant as a Sum over Permutations

In order to explain the polynomial expression for the determinant of an $n \times n$ matrix, we introduce the following notion:

Definition. A *permutation* of length n is a list of numbers $(k_1, \ldots, k_n)$ such that each number from 1 to n appears exactly once in the list.

Example 1. The two permutations of length 2 are $(1, 2)$ and $(2, 1)$. For length 3, we have

$$(1, 2, 3), (1, 3, 2), (2, 1, 3), (2, 3, 1), (3, 1, 2), (3, 2, 1)$$

In general there are $n!$ permutations of length n.

We will use σ to denote a permutation and $\sigma(i)$ the ith entry in σ. Given a permutation σ, we can pick out one entry in each row of an $n \times n$ matrix, by selecting the $\sigma(i)$th entry of row i for example:

$$(3, 1, 2) \rightsquigarrow \begin{bmatrix} & & * \\ * & & \\ & * & \end{bmatrix}$$

For our purposes, one can think of permutations as being patterns like this. The terms in the determinant will be products of these matrix entries, in the above example $a_{13}a_{21}a_{32}$. We denote this product a_σ. The determinant then has the form

$$\det(A) = \sum_\sigma \pm a_\sigma$$

For example, in 2×2, the permutations are $(1,2)$, yielding $a_{11}a_{22}$ and $(2,1)$, yielding $a_{21}a_{12}$, and the determinant is $a_{11}a_{22} - a_{21}a_{12}$.

We now need to explain where the positive and negative signs comes from. For a permutation σ, we denote by $r(\sigma)$ the number of pairs in σ which are "out of order." This is the number of choices of $i < j$ where $\sigma(i) > \sigma(j)$. In terms of the matrix, this is the number of pairs that look like this:

$$\begin{bmatrix} & & & \\ & & * & \\ & & & \\ & * & & \\ & & & \end{bmatrix}$$

The *sign* of σ, $\operatorname{sgn}(\sigma)$ is defined to be $(-1)^{r(\sigma)}$. I can now state the definition of the determinant:

$$\det(A) = \sum_\sigma \operatorname{sgn}(\sigma) a_\sigma$$

Again returning to the case that $n = 2$, our permutations are $(1,2)$ and $(2,1)$. The first has nothing out of order, so $\operatorname{sgn}(1,2) = 1$. The second has one pair out of order, so $\operatorname{sgn}(2,1) = -1$. This gives us the expected formula $a_{11}a_{22} - a_{21}a_{12}$.

This definition is good for explaining the otherwise miraculous properties of the determinant, but we will learn other ways to compute it which are much more efficient in many circumstances.

1.1.1 Some Easy Cases

For the identity matrix, we can say that $\det(I_n) = 1$ for all n. The only permutation for which a_σ is nonzero is $(1,2,3,\dots,n)$. This has sign $+1$, and the corresponding product is the product of the diagonal entries, 1.

More generally, we call a matrix upper triangular if it has zeroes below the diagonal, so looks something like

$$\begin{bmatrix} a_{11} & * & * \\ 0 & a_{22} & * \\ 0 & 0 & a_{33} \end{bmatrix}$$

The determinant of such a matrix is the product of the diagonal entries. Indeed, all permutations except $(1,2,\dots,n)$ pick out something below the diagonal, and so the corresponding term is 0. The same thing is true for lower triangular matrices.

Example 2.

$$\det \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -2 & 1 & 1 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix} = 1 \cdot (-2) \cdot 0 \cdot 1 = 0$$

This matrix is not invertible.

Finally, the determinant of a 1×1 matrix $\begin{bmatrix} a \end{bmatrix}$ is just the number a. The only permutation is (1).

1.2 Dimension Three

Let's see what our definition gives us in the case of 3×3 matrices. Consider a matrix

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

There are 6 permutations, so there will be 6 terms in the determinant. There is a nice visual way to organize these terms and get their signs. Imagine the terms of the matrix extending to the right past the edge, repeating each row:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{matrix} a_{11} & a_{12} & \dots \\ a_{21} & a_{22} & \dots \\ a_{21} & a_{22} & \dots \end{matrix}$$

Now draw 3 lines diagonally down to the right, and 3 up to the right. [Draw picture] Each line picks out 3 terms. The downward lines have sign $+1$ and the upward lines have sign -1. Putting these together, we get

$$a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{31}a_{22}a_{13} - a_{32}a_{23}a_{11} - a_{33}a_{21}a_{12} = \det(A)$$

You can check that these 6 terms correspond to the 6 permutations of length 3, with their corresponding signs.

Example 3. Consider the matrix

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & -1 \\ 1 & 0 & 1 \end{bmatrix}$$

Extending it to the right, we get

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & -1 \\ 1 & 0 & 1 \end{bmatrix} \begin{matrix} 1 & 1 & \dots \\ 0 & 2 & \dots \\ 1 & 0 & \dots \end{matrix}$$

The determinant is then

$$1 \cdot 2 \cdot 1 + 1 \cdot (-1) \cdot 1 + 1 \cdot 0 \cdot 0 - 1 \cdot 2 \cdot 1 - 0 \cdot (-1) \cdot 1 - 1 \cdot 0 \cdot 1$$

$$= 2 - 1 + 0 - 2 - 0 - 0 = -1$$

As this is not zero, we can conclude that the matrix is invertible.

1.3 Cofactor Expansion

For larger matrices, there is a useful recursive method of computing the determinant. We can see an example of this if we look at the formula for the determinant of a 3×3 matrix:

$$\det \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} =$$

$$a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{31}a_{22}a_{13} - a_{32}a_{23}a_{11} - a_{33}a_{21}a_{12}$$

In this expression, we can group terms according to which entry in the first row appears:

$$a_{11}(a_{22}a_{33} - a_{32}a_{23}) - a_{12}(a_{21}a_{33} - a_{31}a_{23}) + a_{13}(a_{21}a_{32} - a_{31}a_{22})$$

The other terms look like 2×2 determinants. In particular, this is

$$a_{11} \det \begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix} - a_{12} \det \begin{bmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{bmatrix} + a_{13} \det \begin{bmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}$$

These 2×2 matrices can be obtained by removing one row and one column from the original matrix, specifically the row and column corresponding to the coefficient outside the determinant. Let's introduce some notation for this. Given a $n \times n$ matrix A, A_{ij} denote the $(n-1) \times (n-1)$ matrix obtained by removing the ith row and jth column. Then in the 3×3 case we can write

$$\det(A) = a_{11} \det A_{11} - a_{12} \det A_{12} + a_{13} \det A_{13}$$

A similar formula works for matrices of any size:

$$\det A = \sum_j (-1)^{1+j} a_{1j} \det A_{1j}$$

This is called a *cofactor expansion* of the determinant. This is a recursive formula, because it reduces the computation of the determinant of an $n \times n$ matrix to the computation of determinants of $(n-1) \times (n-1)$ matrices. This is specifically the expansion across the first row. One can compute the determinant via a cofactor expansion across any row:

$$\det A = \sum_j (-1)^{i+j} a_{ij} \det A_{ij}$$

The only thing that changes is the signs. One can think of the sign of each term in the cofactor expansion as coming from the following pattern in the matrix:

$$\begin{bmatrix} + & - & + & \cdots \\ - & + & - & \\ + & - & + & \\ \vdots & & & \ddots \end{bmatrix}$$

There is also a completely analagous formula that expands across a column as well as a row, using the same pattern of signs.

This method of computing the determinant is especially useful when a column or row contains zeroes, so that many of the terms in the cofactor expansion are just 0.

Example 4. Consider the matrix

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 3 & 1 & 3 & 0 \\ 3 & 2 & 2 & 0 \\ 4 & 0 & 0 & 3 \end{bmatrix}$$

We can expand its determinant as a sum across the bottom row: this is a good choice, because there are only two nonzero terms here:

$$-4 \det \begin{bmatrix} 0 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 2 & 0 \end{bmatrix} + 0 - 0 + 3 \det \begin{bmatrix} 1 & 0 & 1 \\ 3 & 1 & 3 \\ 3 & 2 & 2 \end{bmatrix}$$

We can further expand these 3×3 determinants along rows or columns. For instance, the first can be expanded along the last column, which has only a single nonzero entry, and the second can be expanded along the first row:

$$-4 \left(2 \det \begin{bmatrix} 1 & 3 \\ 2 & 2 \end{bmatrix} + 0 + 0 \right) + 0 - 0 + 3 \left(1 \det \begin{bmatrix} 1 & 3 \\ 2 & 2 \end{bmatrix} - 0 + 1 \det \begin{bmatrix} 3 & 1 \\ 3 & 2 \end{bmatrix} \right)$$

This comes out to

$$-4(2(2-6)) + 3((2-6) + (6-3)) = 32 - 3 = 29$$

Why do these formulas compute the determinant? This can be explained in terms of the definition of the determinant as a sum over permutations. Consider the case of the expansion across the first row. Each permutation selects one element from the first row. We can group permutations according to which element they select. After we have selected an element of the first row, say a_{ij}, we select other entries of the matrix whose row and column are different; these correspond to terms in the determinant of A_{ij}. [Draw picture]

1.4 Determinants and Row Operations

For us, much of the meaning and usefulness of the determinant results from how it is related to row operations. The important properties are recorded below:

1. If B is obtained by swapping two rows of A, then $\det(B) = -\det(A)$.
2. If B is obtained by multiplying a row of A by a number k, then $\det(B) = k\det(A)$.
3. If A_1 and A_2 share all rows except the ith and B is obtained by replacing this ith row with the sum of that of A_1 and A_2, then
$$\det(B) = \det(A_1) + \det(A_2)$$
4. If A has a repeated row, then $\det(A) = 0$.

Each of these properties is a consequence of the definition of determinant as a sum over permutations. As an example, for (1), if we swap two rows, this changes the sign on every term in the determinant. (4) follows from (1), as swapping the repeated rows yields $\det(A) = -\det(A)$.

A useful consequence of (2), (3), and (4) is that the determinant does not change if we add a multiple of one row to another. If B is obtained from A by adding $k \cdot r_i$ to r_j, then $\det(B) = \det(A) + k \cdot \det(C)$, where in C, the row r_i is repeated. Thus $\det(C) = 0$, and so $\det(B) = \det(A)$.

Remark. All the properties listed above are true for operations on columns as well as rows.

With these observations, we can justify the most important property of the determinant.

Proposition. *An $n \times n$ matrix A is invertible if and only if* $\det(A) \neq 0$.

Proof Sketch. If A is invertible, then there is a sequence of row operations bringing it to the identity matrix. Each row operation either multiplies the determinant by -1, if it swaps rows, by a nonzero k, if it scales a row, or does not change it at all, if it adds one row to another. Because the determinant of the identity matrix is $1 \neq 0$, the determinant of the original matrix had to be nonzero as well.

If A is not invertible, then because it is square, there must be a zero row in its reduced row echelon form R. Each permutation selects an entry from this row, so all the products are 0 and so $\det(R) = 0$. As explained above, this means that $\det(A)$ had to have been 0 as well. □

If we keep track of how the determinant changes as we apply row operations, we can use this to compute it, by reducing the matrix to the identity matrix where we know it is 1. In fact, we don't even need to go all the way to the identity matrix; once we get it to a upper triangular matrix, we can compute the determinant by taking the product of the diagonal entries. This is usually the most efficient way to compute determinants of large matrices.

Example 5. Let's compute the determinant

$$\det \begin{bmatrix} 0 & 7 & 5 & 3 \\ 2 & 2 & 4 & 2 \\ 1 & 1 & 2 & -1 \\ 1 & 1 & 1 & 2 \end{bmatrix}$$

using row operations. I will do this in not necessarily the most efficient way, in order to illustrate the method.

$$\det \begin{bmatrix} 0 & 7 & 5 & 3 \\ 2 & 2 & 4 & 2 \\ 1 & 1 & 2 & -1 \\ 1 & 1 & 1 & 2 \end{bmatrix} = -\det \begin{bmatrix} 2 & 2 & 4 & 2 \\ 0 & 7 & 5 & 3 \\ 1 & 1 & 2 & -1 \\ 1 & 1 & 1 & 2 \end{bmatrix} = -2\det \begin{bmatrix} 1 & 1 & 2 & 1 \\ 0 & 7 & 5 & 3 \\ 1 & 1 & 2 & -1 \\ 1 & 1 & 1 & 2 \end{bmatrix} = -2\det \begin{bmatrix} 1 & 1 & 2 & 1 \\ 0 & 7 & 5 & 3 \\ 0 & 0 & 0 & -2 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

$$= 2\det \begin{bmatrix} 1 & 1 & 2 & 1 \\ 0 & 7 & 5 & 3 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & -2 \end{bmatrix} = 2(1 \cdot 7 \cdot (-1) \cdot (-2)) = 28$$

1.5 Multiplicativity of the Determinant

Another useful property of the determinant is its multiplicativity. This means that

$$\det(AB) = \det(A)\det(B)$$

This can be explained in terms of the properties of the determinant we have seen previously, I will put a guided argument for this on the homework.

The multiplicativity of the determinant has some interesting consequences. First of all, if A is invertible, then $\det(A)\det(A^{-1}) = \det(I_n) = 1$, and so $\det(A^{-1}) = \det(A)^{-1}$. Using this, we can also see the following:

Proposition. *If A and B are similar, then* $\det(A) = \det(B)$.

Proof. If $A = SBS^{-1}$, then $\det(A) = \det(S)\det(B)\det(S)^{-1} = \det(B)$. □

As another consequence, we can work out what the determinant of a orthogonal matrix can be. First of all, we have the following.

Proposition. *For any $n \times n$ matrix A,* $\det(A^\top) = \det(A)$.

This can be explained in terms of the definition of the determinant as a sum over permutations: for each term in $\det(A)$, there is a corresponding term in $\det(A^\top)$ with the same value:

$$\begin{bmatrix} & * & \\ & & * \\ * & & \end{bmatrix} \rightsquigarrow \begin{bmatrix} & & * \\ * & & \\ & * & \end{bmatrix}$$

Proposition. *If A is orthogonal,* $\det(A) = 1$ *or* $\det(A) = -1$.

Proof. An orthogonal matrix A satisfies $AA^\top = I_n$, and so $\det(A)^2 = \det(A)\det(A^\top) = 1$. □

Using the next topic of the course, eigenvectors and eigenvalues, we will be able to see that any orthogonal 3×3 matrix with determinant 1 is a rotation about some axis. Those with determinant -1 are this followed by a reflection.

2 Eigenvectors and Diagonalization

Our main application of the determinant will be in *diagonalization.* Given a linear transformation $T : \mathbb{R}^n \to \mathbb{R}^n$, to diagonalize it is to find a coordinate system such that in that coordinate system, T is represented by a diagonal matrix

$$\begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}$$

At this point, you have seen a few examples of this in the homework. Such a representation makes it easier to answer certain questions about T. For instance, T^r will be represented by

$$\begin{bmatrix} \lambda_1^r & & \\ & \ddots & \\ & & \lambda_n^r \end{bmatrix}$$

so this will help us draw conclusions about what happens when we repeatedly apply T.

2.1 Eigenvectors and Eigenvalues

A diagonal matrix

$$D = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}$$

transforms the standard basis vectors $e_1, \dots, e_n$ in a particularly simple way: $De_i = \lambda_i e_i$. The first step of diagonalization will be to identify vectors like this for a given linear transformation.

Definition. For a linear transformation $T : \mathbb{R}^n \to \mathbb{R}^n$ (represented by an $n \times n$ matrix A), a vector v is called an *eigenvector* of T (of A) if $T(v) = \lambda v$ for some scalar λ. Geometrically, this means that $T(v)$ is parallel to v.

This is really only interesting in the case that $v \neq 0$, so from now on "eigenvector" will mean "nonzero eigenvector." In this case, λ is uniquely determined by v, and is called the *eigenvalue* corresponding to v. In general we say that λ is an eigenvalue of A it is the eigenvalue of some nonzero eigenvector.

Example 6. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be reflection across the line $y = x$,

$$\text{span}\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$$

This is represented by the matrix

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Here we have

$$T \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

as vectors on the line are fixed. Thus, this is an eigenvector with eigenvalue 1.

On the other hand

$$T \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix} = - \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

This vector is orthogonal to the line. [Draw picture] It is an eigenvector with eigenvalue -1.

It will be useful to reformulate the condition of being an eigenvector with eigenvalue λ in the following way. The equation

$$Av = \lambda v$$

can be rewritten

$$(A - \lambda I_n)v = 0$$

and so eigenvectors with eigenvalue λ are exactly (nonzero) vectors in $\ker(A - \lambda I_n)$. We call this kernel the λ-*eigenspace* of A, and denote it V_λ.

Question. What is the eigenspace V_0?

Example 7. In the reflection example, the eigenvalues are 1 and -1.

$$V_1 = \text{span}\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$$

$$V_{-1} = \text{span}\left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$$

Remark. When we write V_λ, this depends on the matrix A under consideration.

2.2 The Characteristic Polynomial

From the above reformulation of being an eigenvector gives us a way of finding all the eigenvalues of a matrix and their corresponding eigenvectors. For each λ, $A - \lambda I_n$ is a square matrix, so $\ker(A - \lambda I_n)$ contains a nonzero vector if and only of $A - \lambda I_n$ fails to be invertible. This is true if and only if $\det(A - \lambda I_n) = 0$. Why is this a useful characterization? Well $\det(A - tI_n)$ expands out as a polynomial in t of degree n. We call this the *characteristic polynomial* of A, denoted $p_A(t)$. Plugging in λ, we can say that λ is an eigenvalue if and only if $p_A(\lambda) = 0$.

Proposition. *The eigenvalues of A are exactly the zeroes of the polynomial $p_A(t)$.*

Example 8. For the reflection matrix

$$p_A(t) = \det(A - tI_2) = \det \begin{bmatrix} -t & 1 \\ 1 & -t \end{bmatrix} = t^2 - 1 = (t-1)(t+1)$$

The zeroes of this are 1 and -1, the eigenvalues we saw before.

A consequence of this characterization is that an $n \times n$ matrix has at most n distinct eigenvalues.

Example 9. Consider the rotation matrix

$$R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

This has characteristic polynomial

$$p_A(t) = \det \begin{bmatrix} -t & -1 \\ 1 & -t \end{bmatrix} = t^2 + 1$$

This has no real roots, so the matrix has no (real) eigenvalues. Later, we will introduce and study complex eigenvalues and eigenvectors, which can be applied to matrices like this.

In contrast to this situation, any polynomial of odd degree must have at least one real root. This has the following consequence for matrices.

Proposition. *For n odd, any $n \times n$ matrix has at least one (real) eigenvalue, and so has some nonzero eigenvector.*

Example 10. Consider the matrix

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & -1 & 2 \\ 0 & 0 & -2 \end{bmatrix}$$

The characteristic polynomial is

$$\det \begin{bmatrix} 2-t & 1 & 1 \\ 0 & -1-t & 2 \\ 0 & 0 & -2-t \end{bmatrix} = (2-t)(-1-t)(-2-t)$$

Therefore the eigenvalues are $2, -1, -2$. In general, we can say that the eigenvalues of a upper triangular matrix are the diagonal entries.

2.3 Finding Eigenvectors

Once we have identified the eigenvalues of A a matrix, we can find eigenvectors for each eigenvalue λ by solving the equation $(A - \lambda I_n)v = 0$. In particular, we can find a basis for each nonzero V_λ.

Example 11. Let's consider the upper triangular matrix we looked at above. For the eigenvalue 2, we consider

$$V_2 = \ker \begin{bmatrix} 0 & 1 & 1 \\ 0 & -3 & 2 \\ 0 & 0 & -4 \end{bmatrix}$$

This has reduced row echelon form

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

and so

$$V_2 = \text{span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$$

This is the eigenspace for the eigenvalue 2.

For the eigenvalue -1, we consider

$$V_{-1} = \ker \begin{bmatrix} 3 & 1 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & -1 \end{bmatrix}$$

This has reduced row echelon form

$$\begin{bmatrix} 1 & 1/3 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

and so

$$V_{-1} = \text{span}\left\{ \begin{bmatrix} -1/3 \\ 1 \\ 0 \end{bmatrix} \right\}$$

Sometimes, the eigenspaces will have dimension greater than 1. Consider the matrix

$$H = \begin{bmatrix} 1 & 2 & 1 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

The characteristic polynomial is $(1-t)(-2-t)^2$, and so the eigenvalues are 1 and -2. One can check that

$$V_1 = \ker \begin{bmatrix} 0 & 2 & 1 \\ 0 & -3 & 0 \\ 0 & 0 & -3 \end{bmatrix} = \text{span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$$

On the other hand

$$V_{-2} = \ker \begin{bmatrix} 3 & 2 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \text{span}\left\{ \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix} \right\}$$

has dimension 2.

2.4 Eigenbases and Diagonalization

Suppose that for an $n \times n$ matrix A, we find a basis of $\mathbb{R}^n$ consisting of eigenvectors $v_1, \ldots, v_n$ with eigenvalues $\lambda_1, \ldots, \lambda_n$, so $Av_i = \lambda_i v_i$ for each i. In the coordinate system given by this basis, A is represented by the diagonal matrix

$$D = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}$$

In other words, A takes a general vector $v = x_1v_1 + \cdots + x_nv_n$ to $Av = \lambda_1x_1v_1 + \cdots + \lambda_nx_nv_n$. If we put the vectors in the basis into a matrix S, we can write $D = S^{-1}AS$, which can be rearranged to yield $A = SDS^{-1}$. An expression for A of this form is called a *diagonalization* of A.

Question. Given a diagonalization of A, how can we see if A is invertible? If it is, what is the inverse?

One can think of the diagonalization in the following way. Any vector v can be written $x_1v_1 + \cdots + x_nv_n$. Applying S^{-1}, we extract the coefficients x_i:

$$S^{-1}v = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

Applying D then scales the ith coefficient by λ_i:

$$DS^{-1}v = \begin{bmatrix} \lambda_1x_1 \\ \vdots \\ \lambda_nx_n \end{bmatrix}$$

Applying S then turns this back into a linear combination of the v_i:

$$SDS^{-1}v = \sum_i \lambda_ix_iv_i = Av$$

We will see later that such an expression for A makes it easier to answer certain questions about A.

Example 12. Consider the reflection matrix

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

This has eigenvectors

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

with eigenvalues $\lambda_1 = 1$, $\lambda_2 = -1$. The eigenvectors form a basis for $\mathbb{R}^2$, and so we have the following diagonalization of A:

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}^{-1}$$

2.5 When is a matrix diagonalizable?

We say that a matrix is diagonalizable if it has a diagonalization. From the above remarks, we can say that a matrix is diagonalizable if and only if there is a basis of $\mathbb{R}^n$ consisting of eigenvectors for A. The following proposition makes it more clear what needs to be true for such a basis to exist.

Theorem 1. *Suppose that we choose bases for distinct eigenspaces $V_{\lambda_1}, \ldots, V_{\lambda_k}$, and put them together to get a list of vectors $v_1, \ldots, v_r$. Then this list of vectors is linearly independent.*

I will prove a simpler version of this; on the homework, I will ask you to deduce the above statement from this.

Lemma 1. *Suppose $v_1, \ldots, v_r$ are vectors such that $v_i \in V_{\lambda_i}$ for distinct eigenvalues $\lambda_1, \ldots, \lambda_r$. Then if $v_1 + \cdots + v_r = 0$, it must be that each $v_i = 0$.*

Proof. Suppose $v_1 + \cdots + v_r = 0$. If we apply $A - \lambda_r I_n$ to both sides, we get

$$(\lambda_1 - \lambda_r)v_1 + \cdots + (\lambda_{r-1} - \lambda_r)v_{r-1} = 0$$

Each $\lambda_i - \lambda_r$ is nonzero for $i < r$, and the v_r term has completely gone away. We can keep doing this until we are left with $kv_1 = 0$ for some nonzero k, and so $v_1 = 0$. There is nothing special about this ordering of the vectors, so in fact every $v_i = 0$. ☐

Corollary (of the theorem)**.** *An $n \times n$ matrix A with eigenvalues $\lambda_1, \ldots, \lambda_k$ is diagonalizable if and only if*

$$\dim V_{\lambda_1} + \cdots + \dim V_{\lambda_k} = n$$

By definition, for any eigenvalue λ, $\dim V_\lambda \geq 1$. A consequence of this is that if a matrix A has n distinct eigenvalues, it is diagonalizable. We can get an eigenbasis by choosing one eigenvector for each eigenvalue.

Example 13. We saw previously that the matrix

$$R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

has no (real) eigenvalues, and so it is not diagonalizable. (Later, we will see that this matrix becomes diagonalizable if we work with complex numbers.)

Example 14. Consider the matrix

$$G = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

This has characteristic polynomial

$$p_G(t) = \det \begin{bmatrix} 1-t & 1 \\ 0 & 1-t \end{bmatrix} = (1-t)^2$$

Its only eigenvalue is 1. We can check that

$$V_1 = \ker \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \operatorname{span}\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$$

has dimension 1. Therefore, G is not diagonalizable either. (This one will not be diagonalizable even if we use complex numbers.)

Example 15. Consider the matrix

$$H = \begin{bmatrix} 1 & 2 & 1 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

from before. Recall that for this matrix,

$$V_1 = \operatorname{span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$$

$$V_{-2} = \operatorname{span}\left\{ \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix} \right\}$$

From the theorem above, if we put these vectors together, we get a basis

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix}$$

which diagonalizes H:

$$H = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -3 & -2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -3 & -2 \end{bmatrix}^{-1}$$

2.6 Multiplicities of Eigenvalues

In the previous 2 examples, the characteristic polynomial had repeated roots. Let's introduce some terminology to deal with this.

Definition. For a polynomial $f(t)$ with a root λ, the *multiplicity* of λ as a root of f is the largest m such that $(t-\lambda)^m$ appears as a factor of $f(t)$. If λ is an eigenvalue of a matrix A, its *multiplicity* is its multiplicity as a root of $p_A(t)$.

We will see that this is related to the dimension of the corresponding eigenspaces. However, we first need to understand how eigenvalues of similar matrices are related.

Proposition. *Suppose that A and B are similar. Then $p_A(t) = p_B(t)$.*

Proof. Let $B = S^{-1}AS$. Then

$$B - tI_n = S^{-1}AS - tI_n = S^{-1}AS - S^{-1}tI_nS = S^{-1}(A - tI_n)S$$

This is to say $B - tI_n$ and $A - tI_n$ are similar. Because similar matrices have the same determinant, $p_B(t) = p_A(t)$. □

Corollary. *Similar matrices have the same eigenvalues with the same multiplicities.*

Theorem 2. *If λ is an eigenvalue of A of multiplicity m, then*

$$\dim(V_\lambda) \leq m$$

Proof. Let $v_1, \dots, v_r$ be a basis for V_λ. This can be extended to a basis $v_1, \dots, v_n$ for $\mathbb{R}^n$. Letting

$$S = \begin{bmatrix} \vdots & & \vdots \\ v_1 & \dots & v_n \\ \vdots & & \vdots \end{bmatrix}$$

we have

$$S^{-1}AS = \begin{bmatrix} \lambda & & & * \\ & \ddots & & * \\ & & \lambda & * \\ 0 & \dots & 0 & C \end{bmatrix} = B$$

One can check that $p_A(t) = p_B(t) = (\lambda - t)^r p_C(t)$, and so $m \geq r$. □

We can summarize the criteria for diagonalizability as follows:

1. Given an $n \times n$ matrix A, it has a characteristic polynomial $p_A(t)$. Its eigenvalues $\lambda_1, \dots, \lambda_k$ are the roots of this. Counted with multiplicities $m_1, \dots, m_k$, there are at most n of them; $m_1 + \dots + m_k \leq n$.

2. In order for the matrix to be diagonalizable, we must have $\dim V_{\lambda_1} + \dots + \dim V_{\lambda_k} = n$. In order for this to be true, two things must occur. First, the characteristic polynomial must have enough roots, in the sense that $m_1 + \dots + m_k = n$.

3. Second, for each λ_j, we must have $\dim V_{\lambda_j} = m_j$.

4. If these conditions are met, then we can combine bases of all the V_{λ_j} to get a basis for $\mathbb{R}^n$, this yields a diagonalization $A = SDS^{-1}$ of the original matrix A.

2.7 Complex Eigenvalues

As we have seen, some matrices, like

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

fail to be diagonalizable because the characteristic polynomial, in this case $t^2 + 1$ fails to have enough real roots. On the other hand, this polynomial has complex roots $t = \pm i$. More generally, if we look for roots in $\mathbb{C}$, we always have enough:

Theorem 3 (Fundamental Theorem of Algebra)**.** *A degree n polynomial always has n roots in $\mathbb{C}$, counted with multiplicity.*

To realize these roots as eigenvalues, we need to consider vectors with complex entries as well. We denote the set of such vectors with n entries $\mathbb{C}^n$. Almost all of the results we have for vectors in $\mathbb{R}^n$, linear transformations, matrices, and determinants also apply in $\mathbb{C}^n$, with multiplication by complex scalars replacing multiplication by real scalars. In particular, the notions of subspaces, linear dependence, linear transformations, and dimension work the same way. All the properties of $\mathbb{R}$ that we used, that we can add, multiply, subtract, and divide by nonzero numbers, are also true for $\mathbb{C}$. To clarify when we are working with the complex version of things like dimension, I may write $\dim_{\mathbb{C}}(V)$, for $V \subset \mathbb{C}^n$ a complex subspace, in contrast to $\dim_{\mathbb{R}}(W)$ for $W \subset \mathbb{R}^n$ a real subspace. In particular $\dim_{\mathbb{C}}(\mathbb{C}^1) = 1$, even if you are used to thinking of $\mathbb{C}$ as a 2 dimensional object.

Remark. The following formula is useful for carrying out computations with complex numbers:

$$\frac{1}{a+bi} = \frac{a-bi}{(a+bi)(a-bi)} = \frac{a-bi}{a^2+b^2}$$

There is one exception to the statement that everything works the same way over $\mathbb{C}$, which is the dot product. To see what changes, if we define $v \cdot w$ by the same formula, we can see that

$$\begin{bmatrix} 1 \\ i \end{bmatrix} \cdot \begin{bmatrix} 1 \\ i \end{bmatrix} = 1^2 + i^2 = 0$$

and so we would be inclined to say that this vector has 0 length, even though it is nonzero. There is a way to modify the definition of dot product so that it can be used in the complex world, but for this course we will not need this. In general, we will not mention anything related to length, angle, or orthogonality when we are thinking about vectors in $\mathbb{C}^n$.

With this said, we can apply our previous methods of finding eigenvalues and eigenvectors in the complex world, with the benefit that now our characteristic polynomial will have as many zeroes as we could hope for.

Example 16. Consider again the rotation matrix

$$R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

with characteristic polynomial $p_r(t) = t^2 + 1$. The eigenvalues are the zeroes of this polynomial, $\pm i$. Let's find the corresponding eigenvectors. In other words, we want to find vectors in the kernel of the following matrices:

$$\begin{bmatrix} -i & -1 \\ 1 & -i \end{bmatrix}$$

To find such a vector, we bring the matrix to reduced row echelon form as before:

$$\begin{bmatrix} -i & -1 \\ 1 & -i \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & -i \\ -i & -1 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & -i \\ 0 & 0 \end{bmatrix}$$

This gives us an equation $x = iy$, and so a general element of the kernel has the form

$$\begin{bmatrix} iy \\ y \end{bmatrix} = y \begin{bmatrix} i \\ 1 \end{bmatrix}$$

This is our eigenvector for i. For $-i$, we use the same method:

$$\begin{bmatrix} i & -1 \\ 1 & i \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & i \\ i & -1 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & i \\ 0 & 0 \end{bmatrix}$$

This gives the equation $x = -iy$, and so a general element of the kernel is a scalar multiple of

$$\begin{bmatrix} -i \\ 1 \end{bmatrix}$$

This is our eigenvector for $-i$. Thus we have the following diagonalization of R:

$$R = \begin{bmatrix} i & -i \\ 1 & 1 \end{bmatrix} \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix} \begin{bmatrix} i & -i \\ 1 & 1 \end{bmatrix}^{-1}$$

Allowing for the use of complex numbers, our criteria for diagonalization become simpler:

1. Working in $\mathbb{C}$, the characteristic polynomial $p_A(t)$ always factors completely as

   $$(\lambda_1 - t)^{m_1} \dots (\lambda_k - t)^{m_k}$$

   where $\lambda_i \in \mathbb{C}$ are the eigenvalues, and $m_1 + \cdots + m_k = n$.

2. For each j, the eigenspace $V_{\lambda_j} \subset \mathbb{C}^n$ has dimension $\dim_{\mathbb{C}} V_{\lambda_j} \leq m_j$.

3. If for every eigenvalue, we have $\dim_{\mathbb{C}} V_{\lambda_j} = m_j$, then the matrix is diagonalizable. Otherwise, it is not diagonalizable. In the case that it is diagonalizable, we can find a diagonalization by putting together bases for each V_{λ_j}.